Last week:

Midterm: March 9

- Walks, trails and paths.
- Connected graphs (any two vertices are connected by a path).
  - Distance and geodesics between vertices.
  - Diameter of a graph (greatest distance between two vertices).
- Some common graphs (Paths P_n, Cycles C_n, Cliques K_n).
- Independent set of vertices (induced graph is edgeless).
- Bipartite graphs (the set of vertices can be partitioned with two independent sets).

This week:

- Some common graphs.
- Operations involving graphs (complement, disjoint union, join, cartesian product).
- Multigraphs and digraphs.
- Vertex degrees (handshaking lemma).
- Regular graphs.

Definition. *A bipartite graph* G *with bipartition* $V(G) = X \sqcup Y$ *is called a* ***complete bipartite graph*** *if for every* $x \in X$ *and* $y \in Y$, *we have* $xy \in E(G)$.

Note: • Such a graph is determined by the cardinalities $n = |X|$ and $m = |Y|$, thus denoted $K_{n,m}$.
• A graph of the form $K_{1,m}$ is called a **star**.

$V(G) = X \sqcup Y$ { bipartite: $E(G) \subseteq \{xy : x \in X, y \in Y\}$; complete $E(G) = \{xy : x \in X, y \in Y\}$ ↓ $|E(K_{n,m})| = |X||Y| = n \cdot m$

Example (Complete bipartite graphs).

X:

y:



$K_{2,2}$



$K_{2,3}$



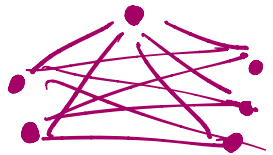
$K_{1,4}$

Definition. • *For* $t \geqslant 2$, *a graph is* t-***partite*** *if* $V(G) = X_1 \sqcup \cdots \sqcup X_t$ *where* $X_1, \ldots, X_t$ *are non-empty independent subsets.*

• *A* t-***partite*** *graph is called a* ***complete t-partite graph*** *if every two vertices in different partite sets are joined by an edge.*

Note: Such complete t-partite graph is determined by the numbers $n_i = |X_i|$, thus denoted $K_{n_1,\ldots,n_t}$.

Example ((Complete) t-partite graphs).

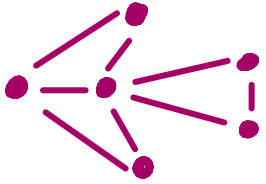


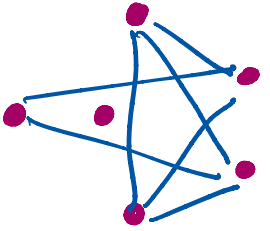
$K_{1,2,3}$

$$|E(K_{n_1,\ldots,n_t})| = n_1(n_2+n_3+\ldots+n_t) + n_2(n_3+\ldots+n_t) + \cdots + n_{t-1}n_t$$
$$= \sum_{j=1}^{t-1} \left(n_j \sum_{\ell=j+1}^{t} n_\ell \right)$$

Definition. *The* ***complement*** $\overline{G}$ *of* G *is the graph with* $V(\overline{G}) = V(G)$ *and an edge* $uv \in E(\overline{G})$ *if and only if* $uv \notin E(G)$.

Example (Graphs and its complements).

G: 

$\overline{G}$: 

$\overline{\overline{G}} = G$

Theorem (1.11). *If* G *is disconnected, then* $\overline{G}$ *is connected.*

Assume G disconnected. WTS $\overline{G}$ connected; i.e: if $u, v \in V(\overline{G})$, then there is a u-v path in $\overline{G}$. Since G NOT connected, it has at least 2 components.

Case 1) u and v belong to dif. components in G, in particular $uv \notin E(G)$, so $uv \in E(\overline{G})$ ($\overline{G}$ has u-v path length 1)

Case 2) u, v belong to the same component. Take w vertex in another component of G. Then $uw \notin E(G)$, $vw \notin E(G)$

Exercise. *Let* G *disconnected. If* $u, v \in V(\overline{G})$, *show that* $d_{\overline{G}}(u, v) = 1$ *or* $d_{\overline{G}}(u, v) = 2$. *What are the possible values for* $\operatorname{diam}(\overline{G})$?

Then $uw \in E(\overline{G})$, $vw \in E(\overline{G})$. Thus (u, w, v) is a u–v path in $\overline{G}$. $\square$

Set-theoretical fact: If X and Y are two sets, one can always form their **disjoint union** $X \sqcup Y$ (a.k.a. coproduct of sets). This is the smallest possible set containing X and Y in a way such that $X \cap Y = \varnothing$.

Definition. *Fix graphs* G *and* H.

(a) The ***disjoint union*** $G \sqcup H$ *is the graph with* $V(G \sqcup H) = V(G) \sqcup V(H)$ *and* $E(G \sqcup H) = E(G) \sqcup E(H)$.

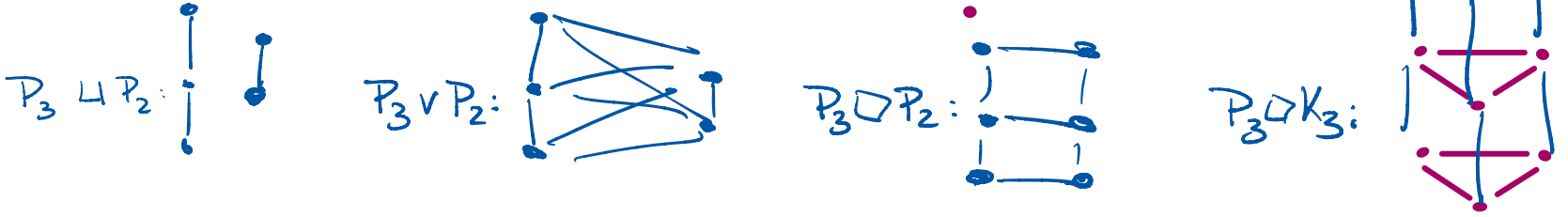
(b) The ***join*** $G \vee H$ *(or* $G + H$*) consists of* $G \sqcup H$ *and all edges joining a vertex of* G *and a vertex of* H*. In other words,*

$$G \vee H = (G \sqcup H) + \{uv : u \in V(G), v \in V(H)\}.$$

(c) The ***Cartesian product*** $G \square H$ *has:*

- *vertices* $V(G \square H) = V(G) \times V(H) = \{(u, v) : u \in V(G), v \in V(H)\}$;
- *an edge between* $(u, v) \neq (x, y)$ *if either* $\begin{cases} u = x \text{ and } vy \in E(H), \text{ or} \\ v = y \text{ and } ux \in E(G). \end{cases}$

Example (Union, join and Cartesian product).

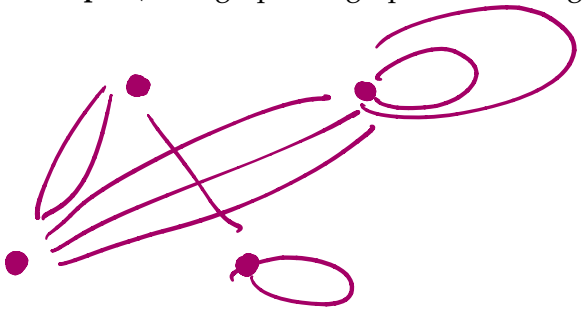


Definition. *(a) A **multigraph** is a graph where the edges can have multiplicities (we call them **multiedges**), and edges of the form vv are also allowed (these are called **loops**).*

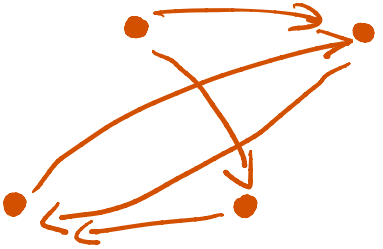
*(b) A **digraph** is a pair $D = (V, E)$ where V is a nonempty set of vertices and $E \subseteq \{(u, v) \in V \times V : u \neq v\}$. The elements of E, called **arcs**, are ordered pairs of different elements of V.*

*(c) An **oriented digraph** is a digraph such that for each pair of vertices u, v, at most one of (u, v) and (v, u) is an arc.*

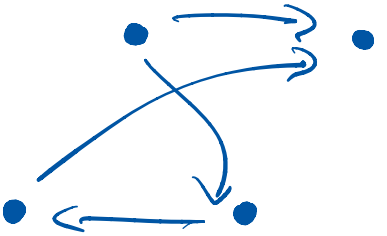
Example (Multigraphs, digraphs, oriented graphs)**.**



Multigraph



Digraph

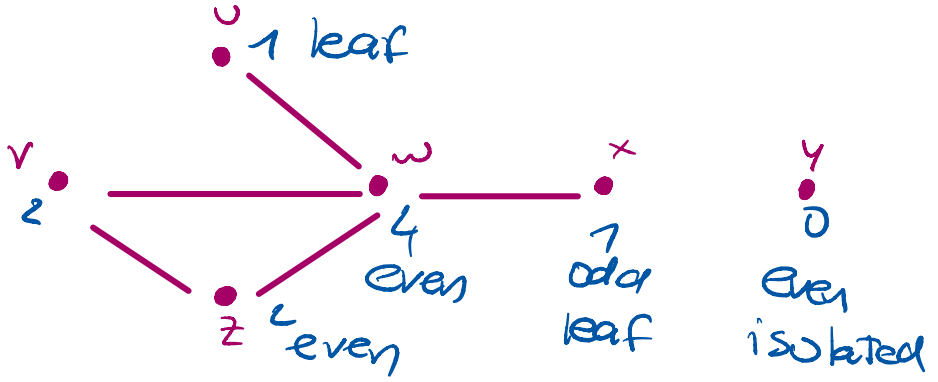


Oriented graph

Definition. *For a vertex v in a graph G, the **degree** of v is $\deg v$ = number of edges incident with v.*

- We say that v is an {**even**, **odd**} **vertex** if $\deg v$ is {even, odd}. 0 is even
- The **neighborhood** of v is the set $N(v) = \{u \in V(G) : uv \in E(G)\}$. Thus $\deg v = |N(v)|$.
- We say that v is an **isolated vertex** if $\deg v = 0$; we say that it is a **leaf** if $\deg v = 1$.
- The **minimum degree** of G is $\delta(G) = \min\{\deg v : v \in V(G)\}$; its **maximum degree** is $\Delta(G) = \max\{\deg v : v \in V(G)\}$.
- If G has n vertices, then $0 \leqslant \delta(G) \leqslant \deg v \leqslant \Delta(G) \leqslant n - 1$.

Example (Degrees, leafs, isolated vertices).



$N(v) = \{w, z\}$

$\delta(G) = 0$

$\Delta(G) = 4$

Theorem ((2.1) The handshaking lemma). *If* G *is a graph, then*

$$\sum_{v \in V(G)} \deg v = 2\,|E(G)|.$$

Proof: when doing the sum $\sum_{v \in V(G)} \deg v$, we are counting each edge two times ∎

Example. *A graph has 14 vertices and 27 edges. The possible values for* $\deg v$ *are 3, 4 or 5, and exactly six vertices have degree 4. How many vertices have degree 3 and how many have degree 5?*

$|E(G)| = 27$

$x = \#$ vertices with deg 3

$y = \#$ vertices with deg $5 = 14 - 6 - x = 8 - x$

$3 \cdot x + 4 \cdot 6 + 5 \cdot y = 2 \cdot 27$, $3 \cdot x + 4 \cdot 6 + 5(8-x) = 54$

$x =$

Corollary (2.3). *Every graph has an even number of odd vertices.*

$V_o = \{v \in V(G) : v \text{ is odd}\}$, $V_E = \{v \in V(G) : v \text{ is even}\}$

$V(G) = V_o \cup V_E$ Then $\sum_{v \in V_o} \deg v + \sum_{v \in V_E} \deg v = 2|E(G)|$

$$\sum_{v \in V_o} \underbrace{\deg v}_{\text{odd}} = 2|E(G)| - \sum_{v \in V_E} \underbrace{\deg v}_{\text{even}}.$$

even

even

Thus $\sum_{v \in V_o} \deg v$ is even, but each summand is odd,

so there must be an even number of summands. □